Replicating LEXAM on German First State Exam–Style Essay Cases to Probe the "Statistical" Face of Legal Reasoning

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Abstract

Large language models (LLMs) increasingly produce fluent legal answers that look like doctrinal reasoning, despite being trained primarily through statistical pattern learning rather than explicit, propositional argument. This tension raises a foundational question for legal theory and legal institutions: do LLMs merely automate peripheral drafting tasks, or do their apparent successes—and characteristic failure modes—reveal that legal reasoning and legal assessment already contain implicit probabilistic and pattern-based elements?

We address this question empirically by replicating the Swiss LEXAM[1] benchmark approach on a new corpus of 200 [2] German, First State Examination–level free-form answer practice cases (*Übungsklausuren*) spanning three difficulty levels. For each case, we rely on the original fact pattern and the official model solution, thereby preserving the exam setting in which legal method is expected to be transparent, disciplined, and normatively constrained. Building on LEXAM's benchmarking logic, we generate LLM answers to these cases and evaluate them along two complementary axes.

First, we introduce a 100-point raw-score rubric tailored to German doctrinal grading practices, designed to capture issue spotting, structured subsumption, doctrinal accuracy, and the normatively relevant weighting of arguments. Second, we report standard NLP evaluation metrics that quantify "closeness" to the model solution: lexical overlap measures, shallow semantic similarity, and textual entailment-based metrics. The central analytic move is to test how far surface and semantic similarity can predict doctrinally meaningful performance under an exam-like scoring regime, including whether these relationships differ systematically across difficulty levels.

Conceptually, the replication is designed to make mismatches productive: where high NLP similarity aligns with high rubric scores, this supports the view that parts of legal reasoning (or at least legal evaluation) are pattern-regular and can be operationalized statistically. Where

[1] Yu Fan et al., *LEXam: Benchmarking Legal Reasoning on 340 Law Exams*, arXiv:2505.12864 (2025), available at https://arxiv.org/abs/2505.12864.

[2] All code and data will be made publicly available for the presentation.

the two diverge—for example, when an answer paraphrases key passages yet omits a decisive doctrinal step, misapplies a standard, or misallocates normative weight—these mismatches help localize the methodological and value-laden dimensions of legal judgment that similarity metrics do not capture.

By extending LEXam-style evaluation to German essay examinations and explicitly relating metric-based similarity to point-based doctrinal assessment, the project contributes (i) a cross-jurisdictional benchmark for free-form legal reasoning, (ii) evidence about what contemporary Language AI can and cannot operationalize in legal method, and (iii) implications for transparency and legitimacy when often private, closed AI systems are introduced into high-stakes legal education and—potentially—legal decision-making institutions.

Disclosure: We used AI for wording and spellchecking.